Fundamental Research 5 (2025) 1034–1046

Contents lists available at ScienceDirect

Fundamental Research

journal homepage: http://www.keaipublishing.com/en/journals/fundamental-research/

Article

Single-cell transcriptome atlas of spontaneous dry age-related macular degeneration in macaques

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A R T I C L E I N F O

Article history:
Received 23 October 2022
Received in revised form 14 February 2023
Accepted 20 February 2023
Available online 25 April 2023

Keywords:
Age-related macular degeneration
Single-cell RNA sequencing
Primate retina
Aging
Müller glia
Ferroptosis

A B S T R A C T

Age-related macular degeneration is the leading cause of irreversible visual impairment in the elderly. It manifests in two forms, wet and dry. However, the mechanisms underlying spontaneous dry age-related macular degeneration (SD-AMD) remain unclear. Herein, we constructed a single-cell retinal transcription atlas in aged non-human primates with SD-AMD. Retinal tissues affected by SD-AMD exhibited a more degenerative and dysfunctional transcriptomic landscape, with global activation of the oxidative stress response and apoptotic signaling pathway. We found two distinct Müller glia subtypes in normal aged and SD-AMD macaques, one exhibiting a photoreceptor-like transcriptome and the other exhibiting a typical Müller glia transcriptome. As SD-AMD progressed, the proportion of photoreceptor-like Müller glial cells decreased, and photoreceptor-function-associated genes were downregulated, indicating weaker Müller glia potential to transit into photoreceptor-like functional states. Microglial cells showed activated features, and the complement system was activated during disease pathogenesis. We also found that the disruption of iron homeostasis and ferroptosis could promote SD-AMD pathogenesis in neural cells. Further experimentation revealed that a ferroptosis inhibitor exerted a profound rescuing effect in SD-AMD mouse models. Based on these results, our study introduces a path toward understanding the pathogenesis of SD-AMD in a non-human primate model at single-cell resolution.

1. Introduction

The retina is a highly organized tissue that senses light and translates optical images to the visual cortex, which is frequently affected by age-related diseases. Specifically, age-related macular degeneration (AMD) remains the main cause of visual dysfunction in elderly individuals, accounting for 8.7% of blindness worldwide [1,2]. Despite its high prevalence and unsatisfactory clinical outcomes, the mechanisms responsible for AMD pathogenesis remain unknown [3]. Comprehensive and thorough evaluation of the mechanisms underlying progressive retinal degeneration is of urgent scientific significance.

Age-related macular degeneration is classified into two forms, wet and dry, based on the presence or absence of retinal neovascularization, respectively. In the wet form, the development of blood vessels invading the retina results in retinal hemorrhage and fibrosis, leading to early, rapid, and severe vision loss [4]. In contrast, dry AMD is characterized by yellow drusen of at least intermediate size, retinal pigment epithelium (RPE) abnormalities, and subretinal reticular pseudo-drusen. While successful disease control by anti-vascular endothelial growth factor (anti-VEGF) drugs is possible for patients with wet AMD, no effective prophylactic or therapeutic measure is available for patients with dry AMD, highlighting a significant medical urgency to investigate its pathogenesis in greater depth.

For decades, researchers have recognized the important roles of RPE inflammation, mitochondrial dysfunction, oxidative stress, and RPE lipofuscin accumulation in dry AMD pathobiology [3]. Genetic studies based on bulk RNA-sequencing have highlighted the importance of the complement system, extracellular matrix remodeling, and lipid abnormalities in AMD pathogenesis [2,5]. However, these complex cell type-specific alterations could be masked by the bulk-tissue-level resolution of bulk RNA-sequencing, underscoring the importance of in-depth transcriptional assessments of dry AMD pathogenesis with resolution at the single-cell level.

Obtaining retinal tissues from disease-free or dry AMD patients is not feasible because of ethical constraints, making it difficult to perform comprehensive studies on dry AMD, especially during its early phase. Thus, non-human primates (NHPs) possessing human-like genetic and physiological features, such as crab-eating macaques (*Macaca fascicu-*

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https://doi.org/10.1016/j.fmre.2023.02.028

laris, also known as cynomolgus monkeys), have become widely used in retina research [6,7]. Moreover, in-depth molecular researches on the cellular classification of the NHP retina based on single-cell RNA sequencing (scRNA-seq) have been achieved, providing a basis for distinguishing different cellular subtypes and further exploring the pathogenesis of non-neovascular AMD [7,8].

In this study, we selected aged cynomolgus monkeys with or without spontaneous dry AMD (SD-AMD) and constructed a single-cell retinal transcription atlas of the retinas of both groups. By characterizing distinct retinal cell subtypes and transcriptomic signatures, we revealed common transcriptomic alterations associated with increased oxidative stress, apoptotic signaling pathway activation, and degenerative tendencies, as well as cell type-specific alterations in SD-AMD. Our study provides a roadmap toward understanding the pathogenesis of SD-AMD based on an NHP model at single-cell resolution.

2. Materials and methods/experiment

2.1. Materials

Pentobarbital sodium, Yaxin Pharmaceutical Co. Ltd.
Papain, Sigma-Aldrich, P3125
Dnase I, Sigma-Aldrich, AMPD1
DMEM, Gibco, 11995081
PBS, HyClone, SH30256.01B
BSA, Roche, 10735078001
Single Cell 3′ Reagent Kits v3, 10xGenomics, 1000158
Qubit^{TM} dsDNA HS Assay Kit, Thermo Fisher Scientific, Q32851
Highly sensitive DNA test Kit, Agilent, 5067-4626
Cell Strainer, BD, 352350
Automated Cell Counter, Countstar, 120005
Trypan Blue, Thermo Fisher Scientific, 15250061
Centrifuge tube, Corning, NJ222402100
Dynabeads^{TM} MyOne^{TM} Silane, Thermo Fisher Scientific, 37002D
SPRIselect, Beckman, B23318
Sodium iodate, Sigma-Aldrich, 71702-25G
Ferrostatin-1, Sigma-Aldrich, SML0583-5MG

2.2. Methods

2.2.1. Screening of SD-AMD in cynomolgus macaques

A series of eye examinations was performed on more than 600 cynomolgus macaque monkeys in both eyes to screen for the AMD models, including intraocular pressures (IOP) (Tono-Vet), slit lamp exams, fundus imaging, OCT, etc. All animals were purchased from Huazhen Animal Feeding Center (Guangdong, China). The study was approved by the Ethical Committee of the Zhongshan Ophthalmic Center (Permit Number: SYXK (YUE) 2018-0189). All animal experiments were consistent with the Animal Welfare Act in accordance with the Standing Committee on Ethics in China (State Scientific and Technological Commission of China).

Inclusion criteria: intraocular pressure (IOP) in the interval of 10–21 mmHg, normal anterior chamber, a certain number of drusen in the macular area with fundus photography and located in retinal pigment epithelial layer with OCT. Eligible monkeys were included in the experimental group. Besides, those with additional ocular diseases or systemic diseases were excluded. Moreover, the monkeys were re-examined 2 weeks and 1 months after the initial tests to confirm the AMD diagnosis. Finally, we have identified 6 aged monkeys as spontaneous AMD models. Another 4 healthy aged monkeys were selected as the control group.

2.2.2. Extraction of the retina

Six eyes of the 6 AMD monkeys (female, 19–22 years of age) and 4 eyes of the 4 healthy aged monkeys (female, 20–21 years of age) were used in this study. The animals were euthanized by an overdose of pentobarbital sodium (50 mg/kg, Yaxin Pharmaceutical Co. Ltd., China) via intravenous injection. Eyes were enucleated, and transported to the laboratory on ice within 8 h after death. Muscles and conjunctiva fascia of the eyeball were removed under a microscope. An incision around the pars plana was first performed to remove the anterior segment including cornea, lens, iris, and ciliary body. The retinal layer (retinal tissues 6 mm diameters centored on the foveal pit) was dissected from the choroid.

2.2.3. Single cell dissociation

Retina tissues were placed in the tissue preservation solution and transported to the laboratory at 4 °C. Then, the samples were removed from the tissue preservation solution and transferred to a pre-cooled 5 ml centrifuge tube. Tissues were incubated in 20 units/ml papain with 0.005% DNase at 37 °C on a shaker, and gently mechanically dissociated with a pipette for 10 min. Then, digested cells were resuspended in 0.04% nonacetylated BSA and filtered through a 70 μm filter to achieve the maximum number of cells. Viability and cell number were assessed via Trypan Blue (ThermoFisher, USA) and an automatic cell counter (Countstar, China). By adding PBS (with 0.4% BSA), cell concentration was adjusted to meet the requirements of barcoding and library construction (around 1.0×10^6 cells/ml). Finally, retinal samples were subjected to library construction.

2.2.4. Barcoding and library construction

We calculated and prepared proper volumes of each sample, aiming at capture of 10,000 cells, and loaded the samples on the 10X Genomics single-cell-A-chip. Each sample was subjected to droplet generation, heat-sealing, reverse transcription (Veriti 96-well thermal cycler, Thermo Fisher, USA), and cDNA recovery provided by 10X Genomics as recommended in the user guide. Then, we amplified the purified cDNA for 12 cycles, and cleaned it up with SPRIselect beads (Beckman, USA). The concentration of cDNA was calculated on a Bioanalyzer (Agilent Technologies, USA). We prepared cDNA libraries with 10X Genomics Single Cell 3′ Reagent Kits v3 according to user guide.

2.2.5. scRNA-seq data processing and quality control

FastQC software was utilized to test library quality. Reads were mapped to the *Macaca fascicularis* reference with CellRanger (Version 4.0.2, 10X Genomics), followed by quantification of reads for each sample using default parameters. The *Macaca fascicularis* reference was built following instruction recommended by 10X Genomics. For quality control, we utilized the Seurat package (version 4.0), and filtered out low-quality cells (detected mitochondrial genes greater than 5%, or detected genes fewer than 200 or greater than 2,500).

2.2.6. scRNA-seq analysis

We used the Seurat package for sample integration, normalization, dimensionality reduction, clustering and DEG analysis following the Seurat integration vignette with default parameters. The top 2,000 most variable genes defined by the 'FindVariableFeatures()' script for each sample were used to select variable genes across samples. Unsupervised clustering was achieved by the 'FindClusters()' script at an appropriate resolution to identify significant clusters. We determined DEGs for each cluster with the 'FindAllMarkers()' function, and DEGs for different groups of animals with 'FindMarkers()' funciton. Adjust P values were generated based on Bonferroni correction utilizing all features in the datasets. DEGs were defined with adjust P value < 0.05, and |Log2 fold-change| > 0.25). R package Monocle3 was utilized to construct pseudo-time differentiation trajectory following the guided tutorial. The single-cell sequencing data have been deposited at GSA with the project number PRJCA011088.

2.2.7. GO analysis and senescence-associated secretory phenotype (SASP) score calculation

We utilized Metascape (www.metascape.org) to perform GO analysis. Visualization of GO analysis was performed with R package ggplot2 (version 3.3.0). The SASP score is calculated as the average expression level of SASP-related genes in each cell of a given cell type. Gene expression levels were transformed into log2 TPM based on raw TPM from scRNA seq data, and SASP-related gene list was provided by Wang et al. [6].

2.2.8. Cell-cell communication analysis

We calculated the intercellular communication with CellPhoneDB software (version 1.1.0) (www.cellphonedb.org). Ligand-receptor pairs expressed in at least 10% cells of the same type were included and analyzed, with the expression of both ligand and receptor being detectable. The average expression of each pair in different groups of macaques was compared in a cell-type-specific matter. Interaction pairs with $p < 0.05$ were selected for further analysis.

2.2.9. Dry AMD mouse model induction and treatment

Guangzhou Animal Testing Center offered 6–8 weeks old female C57BL/6 J mice. To induce dry AMD, mice were injected into the tail vein with 35 mg/kg $NaIO_3$ in 100 uL 0.9% NaCl. The animals were randomly divided. Animal experiments followed the ARVO Statement and institutional policies (Sun Yat-Sen University). Mice in anti-ferroptosis group received Fer-1 injection (0.8 mg/kg) via the tail vein 1 hour after model induction.

2.2.10. Electroretinography (ERG) test

ERG test was performed on dry AMD mouse models 14 days after model induction. After maintenance in darkness for one day to dilate pupils, mice were anesthetized, and flash ERGs were captured from the left eye under red light with a constant temperature of mice. The gold-ring electrode (Mayo, Aichi, Japan) contacted the cornea, the reference electrode (Nihon Kohden, Tokyo, Japan) was inserted into the tongue, and the ground electrode (Nihon Kohden) was inserted subcutaneously near the tail. The amplitude of the a-wave (baseline to a-wave trough) and b-wave (a-wave trough to b-wave trough) was measured.

2.2.11. Histological analysis

To obtain retinal tissue, we enucleated and immersed the eyes in 4% paraformaldehyde (PFA) for at least 24 h at 48 °C. We prepared 5 um paraffin-embedded sections of retinal tissues, followed the standard workflow, and stained the samples with hematoxylin and eosin (H&E). A fluorescent microscope (BZ-9000; Keyence, Osaka, Japan) was used to take images.

2.2.12. Statistical analysis

Data analysis was performed with GraphPad Prism Software. The values are represented as the mean ± SD. Unpaired, two-tailed Student's *t*-test was utilized in this study. P values over 0.05 were considered as not significant, NS; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; and ****, $p < 0.0001$.

3. Results

3.1. Establishment of the single-cell atlas of normal aged retinas and SD-AMD retinas

To investigate transcriptomic alterations during SD-AMD pathogenesis, we selected aged cynomolgus monkeys (19–22 years old) with or without SD-AMD, which was determined by fundus photography and optical coherence tomography (OCT) results. Based on previous research, we identified multiple scattered yellow drusen in the posterior pole, accompanied by RPE abnormalities, in SD-AMD monkeys (Figs. 1a–c and S1a, b) [9]. We then established a single-cell transcription atlas from the retinas of six macaques with dry AMD and those of four normal aged controls. We dissected the neural retina of the macaques and performed single-cell RNA-sequencing on them (Fig. 1a). After quality control, we selected approximately 30,000 high-quality retinal cells for further analysis (Fig. 1d, e). Nine different cell types were identified via unsupervised clustering, each with specific transcriptional features. No differences in cluster distribution or marker gene expression were observed between the diseased and control groups of monkeys (Fig. 1f, S1c–e). The cell types identified included rod cells, cone cells, bipolar cells (BCs), amacrine cells (ACs), horizontal cells (HCs), retinal ganglion cells (RGCs), Müller glial cells, microglia from the neural retinal layer, and RPE cells from the choroid layer. We next explored changes in cell type proportions between aged cynomolgus macaques with SD-AMD and normal aged controls (Fig. 1g). There was a decrease in the proportions of cone, rod, and RPE cells in SD-AMD individuals, implying that those cell types were involved in SD-AMD pathogenesis and progression.

3.2. Effect of SD-AMD on global transcriptomic alterations

We next explored SD-AMD-associated alterations in gene expression in different cell types, identifying differentially expressed genes (DEGs) by comparing gene expression patterns SD-AMD retinas to controls. Notably, we identified more upregulated DEGs in BCs, rod cells, and Müller glial cells, and more downregulated DEGs in HCs, RGCs, and BCs, compared to the expression changes identified in other cell types in AMD monkeys. This indicated that those cell types were more likely to be affected by SD-AMD (Fig. 1h).

According to prior research, increasing age is robustly and consistently associated with AMD in humans, and over 10% of people older than 80 years suffered from late AMD [10]. Moreover, researchers have highlighted the contribution of the senescence-associated secretory phenotype (SASP), which represents an age-related, low-grade inflammatory microenvironment produced by senescent cells, to the onset of age-associated diseases [11]. By evaluating and comparing the senescent state of different retinal cell types from SD-AMD-affected and normal aged animals, we discovered that those with SD-AMD exhibited higher SASP scores in the rod, cones AC, BC, RPE, and Müller glial cell subtypes. Our results suggest that these cell types exhibit more pronounced senescence-associated alterations during early AMD pathogenesis (Fig. 1i).

3.3. Two distinct subtypes of Müller glia were identified in normal aged and SD-AMD macaques

Müller glia, the main glial components of the retina, have been shown to participate in various retinal biological processes, including providing multiple neurotrophic/growth factors, regulating inflammatory responses, and eventually mediating neuronal survival of the retina [12]. Accumulating evidence has demonstrated the remarkable plasticity and multipotent regenerative potential of Müller cells. To further explore the transcriptional alterations and define the potential roles of Müller cells in SD-AMD pathogenesis, we selected and re-clustered Müller glial cells with unsupervised clustering (Figs. 2a, b and S2a–d). Subsequently, we identified two distinct subtypes of Müller glial cells with specific gene expression landscapes: photoreceptor-like (PR-like) and typical Müller glial cells (Fig. 2c). Both subtypes expressed high levels of Müller glia marker genes (Fig. 2d). The PR-like subtype, representing about 10% of the Müller glial cells, was characterized by higher expression of photoreceptor marker genes, including *RHO, SAG, GNAT1*, and *ROM1*, compared to the other subtype. Further, the expression of these genes that was almost comparable to that in the photoreceptors (Fig. 2c, e and S2e). The other subtype showed a more "typical" transcriptome landscape, characterized by higher levels of *AQP4, CD9*, and *CD81* (Fig. 2c and S2a–d). Trajectory analysis also confirmed that Müller glial cells transit into different functional states, with a distinct branch identified in the PR-like subtype (Fig. 2f). Functional enrichment analysis of the upregulated genes in the PR-like subtype showed enrichment in light sensitivity-associated categories, including "phototransduction

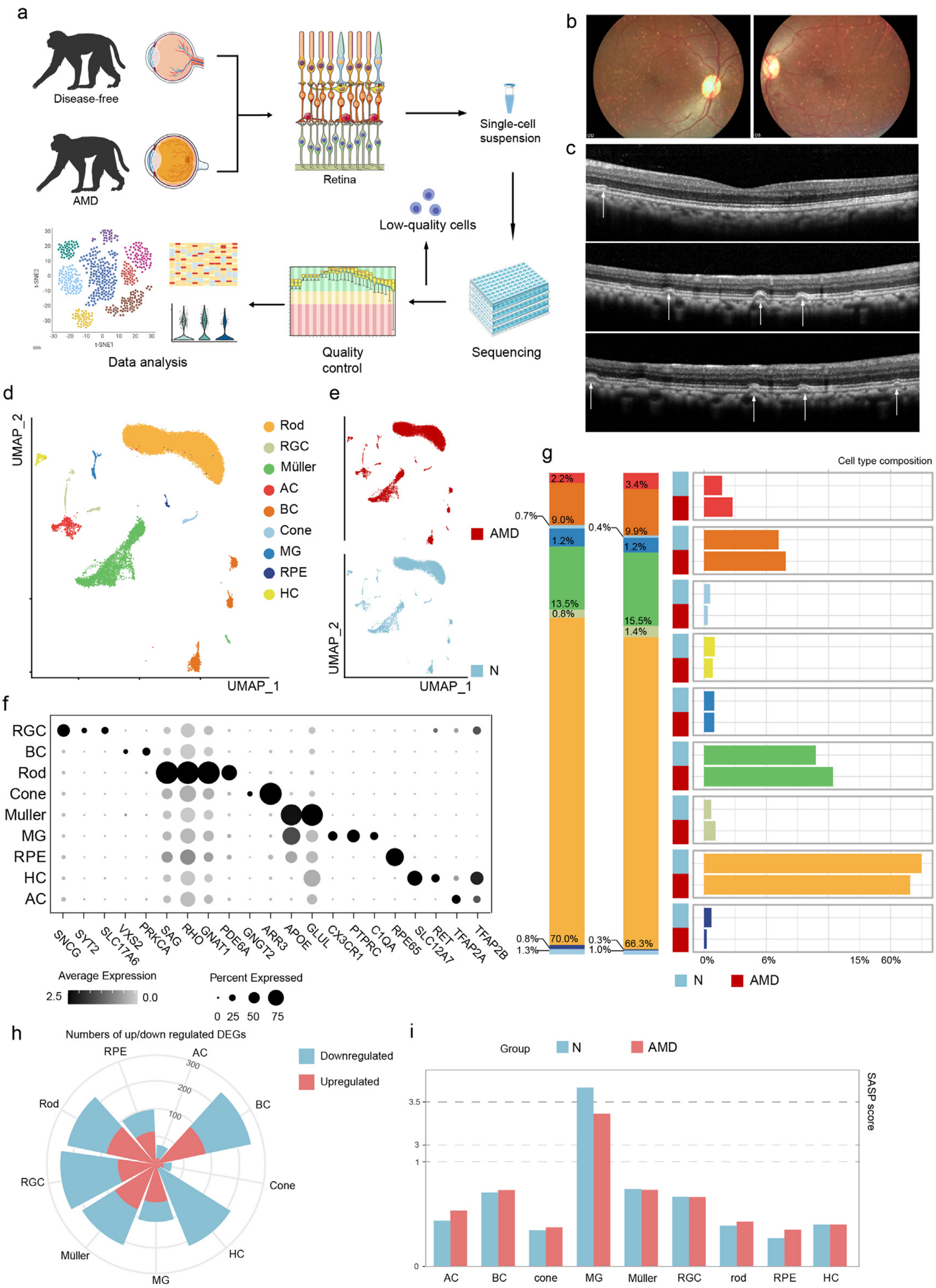

Fig. 1. Study design and classification of retinal cells. (a) Research design. Retinal tissues of spontaneous dry AMD and normal aged crab-eating macaques were studied. Five individual samples respectively from two groups are used to be sequenced. IRBP, interphotoreceptor retinoid-binding protein. PTX, pertussis toxin. AMD, age-related macular degeneration. (b) Fundus photography of dry AMD macaques. (c) OCT images of dry AMD macaques. (d) UMAP embeddings of retinal cells. Rod, rod cells; RGC, retinal ganglion cells; Müller, Müller cells; AC, amacrine cells; BC, bipolar cells; Cone, cone cells; MG, microglia; RPE, retinal pigment epithelium cells; HC, horizontal cells. (e) UMAP embeddings of retinal cells colored by AMD or normal control groups. (f) Dot plots of marker genes for each type of cells. (g) Bar plots of proportions of each subtype of retinal cells in AMD and normal group. (h) Rose diagrams of numbers of upregulated/downregulated DEGs in each type of cells. (i) Bar plots of SASP scores of each type of retinal cells in dry AMD and normal group. N, normal aged controls; AMD, spontaneous dry AMD macaques.

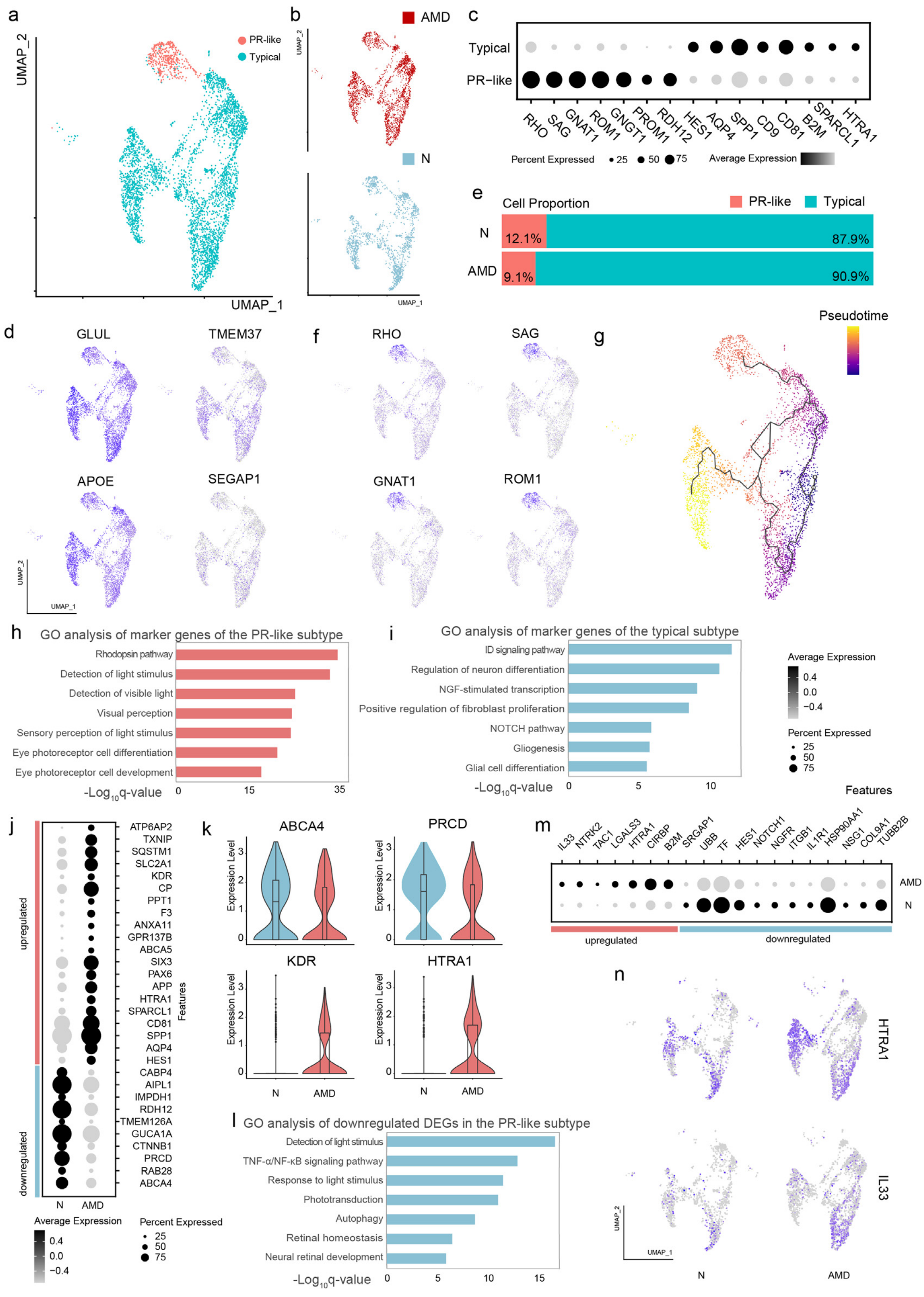

Fig. 2. Identification of two distinct subtypes of Müller glia cells and cell-type-specific alterations in SD-AMD macaques. (a) UMAP embeddings of Müller glial cells colored by subtypes. PR-like, the photoreceptor-like subtype; Typical, the typical subtype. (b) UMAP embeddings of Müller glial cells colored by cell origins. N, normal aged controls; AMD, SD-AMD macaques. (c) Dot plots of marker genes of Müller glia subtypes. (d) UMAP embeddings of Müller glial cells colored by expression of Müller glia marker genes. (e) Bar plots of cell proportions of two Müller glia subtypes of two groups of macaques. (f) UMAP embeddings of Müller glial cells colored by expression of the PR-like subtype marker genes. (g) UMAP embeddings of Müller glial cells colored by pseudo-time differentiation trajectory. Black line indicates the computational differentiation trajectory of Müller glial cells. Number shows the origin plot. (h-i) Bar plots of GO analysis of marker genes of the PR-like (h) or typical (i) subtype. (j) Dot plots of upregulated/downregulated DEGs of the PR-like subtype comparing two groups of macaques. (k) Violin plots of *ABCA4, PRCD, KDR* and *HTRA1* in the PR-like subtype in two groups of macaques. (l) Bar plots of GO analysis of downregulated DEGs in the PR-like Müller glial subtype. (m) Dot plots of upregulated/downregulated DEGs of the typical subtype comparing two groups of macaques. (n) UMAP embeddings of Müller glial cells from two groups of macaques colored by expression of HTRA1 and IL33.

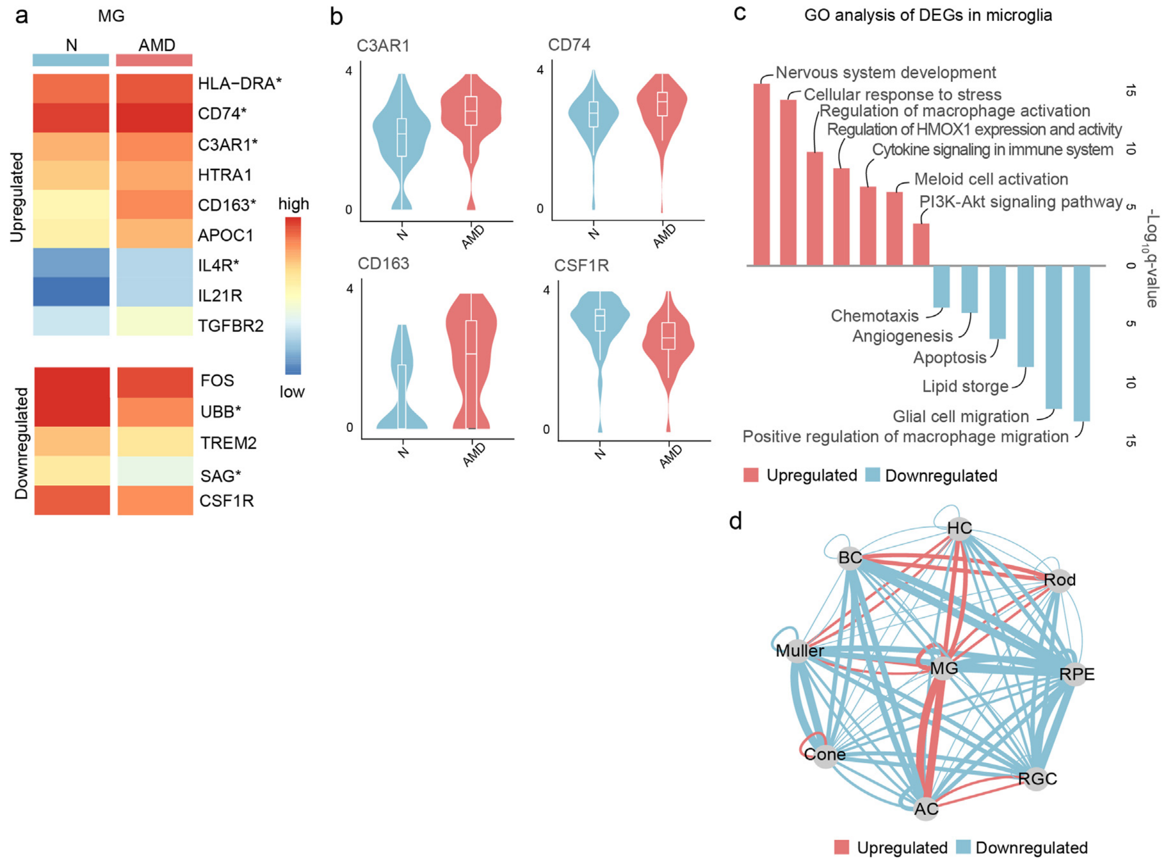

Fig. 3. Microglia are activated in monkeys with spontaneous dry AMD. (a) Heatmap of the expression of DEGs in microglia from two groups of macaques. (b) Violin plots of *C3AR1, CD74, CD163* and *CSF1R* in microglia from two groups of macaques. (c) Bar plots of GO analysis of DEGs in microglia. (d) Networks of the alteration of the intercellular interactions between nine different types of cells comparing spontaneous dry AMD macaques and normal aged controls. The lines indicate intercellular communication, the thickness of which corresponded to the number of intercellular ligand-receptor pairs. Abbreviations for each type of cells are the same as that in Fig. 1.

pathway", "detection of light stimulus", and "visual perception", which further supported our theories (Fig. 2g).

The marker genes identified in the typical subtype were enriched for categories related to glia function, and associated pathways, including "NGF-activated", "NOTCH pathway", and "gliogenesis" (Fig. 2h). Our results indicate that, in vivo, NHP Müller glia comprise two subtypes with potentially distinct biological functions. While most of the Müller glial cells exhibited the more typical transcriptional features of glial cells, others tended to transit into a distinct functional state, expressing high levels of light-sensitivity-associated genes in vivo.

3.4. Transcriptomic alterations of two Müller glial subtypes between SD-AMD and normal aged controls

We sought to explore the transcriptional differences in Müller glial cells between SD-AMD and normal aged macaques. By comparing cell composition, we found that the proportion of the PR-like subtype was reduced in the SD-AMD group, which suggested weaker transitioning ability of Müller glial cells into PR-like states in macaques with SD-AMD (Fig. 4e). We then explored the transcriptional differences between the two Müller glial subtypes in the two groups of macaques. We found that in the PR-like subtype, multiple genes associated with photoreceptor function (*CABP4, AIPL1, ABCA4*, and *RAB28*) and retinal disease (*IMPDH1, PRCD, TMEM126A*, and *GUCA1A*) were downregulated in SD-AMD macaques (Figs. 2i and S2f). Our results indicated weaker potential of Müller glial cells to transfer into photoreceptor-related functional states under SD-AMD states. Specifically, we identified downregulation of *ABCA4* and *RAB28* to be correlated with photoreceptor dysfunction, *PRCD* to be essential in photoreceptor morphogenesis, and *CTNNB1* to participate in retinal development (Fig. 2j). Functional enrichment of downregulated DEGs indicated enrichment in several categories, including "detection of light stimulus", "phototransduction", "retinal homeostasis", and "neural retinal development" (Fig. 2k). Regarding upregulated genes in the PR-like subtype, we found upregulation of *HTRA1* to be highly associated with AMD susceptibility. The *KDR* gene, which encodes VEGFR2, was also upregulated in SD-AMD macaques, suggesting early neovascularization-associated alterations (Fig. 2j).

Regarding the typical subtype, we also identified notable transcriptomic changes. Compared to normal aged macaques, SD-AMD macaques exhibited upregulated *IL33,* which encodes key regulators of inflammation and photoreceptor degeneration, as well as upregulated *HTRA1* (Fig. 2l, i). We identified downregulated expression of the transcription factor *NOTCH1*, which is involved in one of the key pathways regulating Müller glia-based retinal regeneration (Fig. 2i). Furthermore, *NGFR,* which encodes a neural growth factor receptor, and *IL1R1,* which encodes an interleukin-1 receptor, were downregulated in SD-AMD macaques (Figs. 2i and S2g). Downregulated DEGs of the typical subtype were enriched for the categories of "neural stem cell population maintenance", "canonical and non-canonical NOTCH signaling", and "neuron system development" (Fig. S2h).

3.5. Microglia are activated in SD-AMD monkeys

As the primary resident immune cell type, microglia participate in neurodegeneration [13]. Thus, we first explored the role of microglia in retinal degeneration, unveiling their unique potential in SD-AMD pathogenesis. In an AMD state, microglia exhibited a relatively activated landscape, as suggested by the upregulated expression of activation-associated genes (*HLA-DRA, CD74, C3AR1, CD163*, and *IL4R*) and the downregulation of degeneration-associated pathways (*SAG*) (Fig. 3a) [14–17]. We identified upregulated expression of the complement receptor gene *C3AR1*, which is closely associated with complement system activation and nerve degeneration [18]. We also found an increase in the expression of *IL4R, CD163*, and *CD74*, indicating that microglia differentiate into M2 phenotypes, which possess weaker phagocytosis capabilities (Fig. 3b and S3a) [14]. We further identified downregulation of CSF1R expression, which could account for the diminished chemotaxis abilities and population depletion observed in the microglia of individuals affected with SD-AMD (Fig. 3b) [19].

Gene Ontology (GO) analysis further supported our hypothesis of microglia activation, with upregulated DEGs being enriched for the categories of "regulation of macrophage activation", "cytokine signaling in immune system", and "myeloid cell activation" (Fig. 3c). Corresponding to previous research, we also found that downregulated DEGs were enriched for chemotaxis-related processes, including "positive regulation of macrophage migration", "glial cell migration", and "chemotaxis", as well as "angiogenesis", further suggesting their protective roles in SD-AMD pathogenesis (Fig. 3c).

To better characterize the alterations in microglia, we investigated the intercellular communication between microglia and other retinal cells by quantifying the number of ligand-receptor pairs in SD-AMD monkeys and normal aged controls (Fig. S3b). Compared to the controls, the retinal tissues of SD-AMD monkeys exhibited enhanced cell–cell interactions between microglia and other cells, implying microglia activation in SD-AMD pathogenesis (Fig. 3d).

3.6. Photoreceptors show distinct transcriptional alterations during SD-AMD

Age-related photoreceptor loss and dysfunctional changes in rod photoreceptors have been extensively reported in aging retinal tissues [20]. Moreover, one of the earliest complaints of patients with AMD is visual difficulties in dark adaptation [4]. Combined with the previously identified higher SASP scores in photoreceptors, we hypothesized that rod photoreceptors tend to switch to a more dysfunctional, degenerative, and senescent state in SD-AMD, compared to rod cells in controls. Our data showed that rod cells exhibited profound dysfunctional changes in their transcriptomes, along with decreased rod cell proportions, as previously mentioned (Fig. 4a). We also observed downregulated expression of functional genes in rods (*RDH12, PRPH2, CABP4, CABP2*, and *ROM1*), with downregulated DEGs being enriched for "visual perception", "rhodopsin pathway", "photoreceptor development", and "detection of light stimulus", based on GO analysis (Figs. 4b–d and S3b). Categories such as "photoreceptor development" and "eye development" were also enriched, which indicated rod photoreceptor degeneration. These results suggest profound functional damage as well as degenerative and senescence-associated changes in rods during SD-AMD pathogenesis.

Moreover, we found upregulated expression of the oxidative stress-associated genes *DUSP1* and *ROCK2*, which encode a major regulator of axonal degeneration (Figs. 4e and S3c). The expression levels of several transcription factors, such as *FOS* and *CIRBP*, were also upregulated in SD-AMD monkeys (Figs. 4e and S3c). Furthermore, we found enhancement of "cellular response to lipid", "response to oxidative stress", "RHOA pathway", and "response to cAMP" via GO analysis, suggesting profound metabolic abnormalities in rods during AMD pathogenesis (Fig. 4d).

However, similar dysregulated function-associated features were not observed in cone cells, indicating cell type-specific transcriptomic alterations between the different types of photoreceptors. Rather, we found downregulated expression of the cell cycle-related genes *GINS2* and *RTN*, as well as downregulation of *ABCC4*, which is associated with cAMP transport [21]. This may explain the decrease in cone cell dysfunction and depletion observed in SD-AMD macaques (Fig. 4f).

3.7. RPE cells exhibit neovascularization-associated features in SD-AMD macaques

The RPE is primarily responsible for photoreceptor health and function, forming a critical barrier between the retina and systemic circulation [22]. Dysfunction of the RPE has been widely implicated in AMD pathogenesis, contributing to the downstream formation of drusen, inflammation, and ultimately neovascularization. We identified upregulated expression of neovascularization-associated genes in RPE cells in SD-AMD retinas, including *RAC1, PTN, CLIC4, EGFL7*, and *TGFBR3* (Fig. 5a, b) [23–25]. Rac1 activation was found to be important for endothelial cell invasion into the neural retina [23]. This suggested that the RPE exhibits angiogenesis-related transcriptomic alterations even in early SD-AMD, which could differentiate disease progression between the dry and the wet form. Functional enrichment analysis indicated the corresponding upregulation of "angiogenesis", "blood vessel development", and "VEGFA-VEGFR2 signaling pathway", with apoptosis-related pathways also being enhanced (Fig. 5c). Moreover, GO analysis revealed that downregulated genes in SD-AMD were enriched for "lipid storage", "lipid-lipid localization", "eye development", "ion homeostasis", and "cation homeostasis", suggesting disordered lipid metabolism and ion homeostasis (Fig. 5d). Furthermore, we identified upregulated expression of *STRA6*, a key regulator that mediates vitamin A transport into the retina (Fig. 5e).

3.8. Cell type-specific alterations in neural cells indicated the contribution of ferroptosis in SD-AMD

To categorize DEGs into those shared by multiple neural cells and those which are cell type-specific, we generated Venn diagrams of DEGs among the four types of neural cells (ACs, BCs, HCs, and RGCs; Fig. 6a). The levels of *RBM3* and *CIRBP*, both of which are reportedly associated with cellular response to oxidative stress, were commonly upregulated, suggesting generally elevated oxidative stress in neural cells (Fig. 6b, c) [26]. In contrast, commonly downregulated genes were involved in neural growth (*NCDN* and *APLP1*) (Fig. 5a, d, e) [27]. Interestingly, APLP1 reportedly participates in Alzheimer's disease (AD) pathogenesis and neural cell survival, indicating potential similarity between the mechanisms underlying AD and AMD (Fig. 6e) [28]. Further, GO analysis also supported our hypothesis, with downregulated genes (freq $\geq$ 2) being enriched for "iron ion homeostasis", "visual phototransduction", and "nervous system development", whereas upregulated genes were enriched for "autophagy" and "phagosome" terms (Fig. 6f). Thus, the disruption of iron homeostasis could be a contributor to SD-AMD-associated neural cell degeneration.

We next sought to define the cell type specificity of these DEGs across four types of neural cells. Based on GO analysis, upregulated genes were enriched for the response to stress, oxidative stress, and the apoptotic signaling pathway. We also identified similar enrichment for neuron development and the iron delivery-correlated pathway among downregulated genes (Fig. 7a). We found that upregulated DEGs were enriched for ferroptosis of RGCs, suggesting that ferroptosis could be implicated in AMD pathogenesis (Fig. 7a).

Based on our results, we hypothesized that ferroptosis plays a central role in dry AMD pathogenesis. To validate our hypothesis, we constructed mouse models of dry AMD, treated them with a ferroptosis inhibitor, and compared the treated and untreated mouse models to normal controls [29]. We observed subretinal, extracellular deposits in Bruch's membrane as well as the global degeneration of RPE cells and photoreceptors in dry AMD mouse models (Fig. 7b). Furthermore, the ferroptosis inhibitor strongly reversed these effects, mitigating the degeneration of retinal photoreceptor cells (Fig. 7c). We also investigated the overall retinal function in dry AMD mouse models by performing

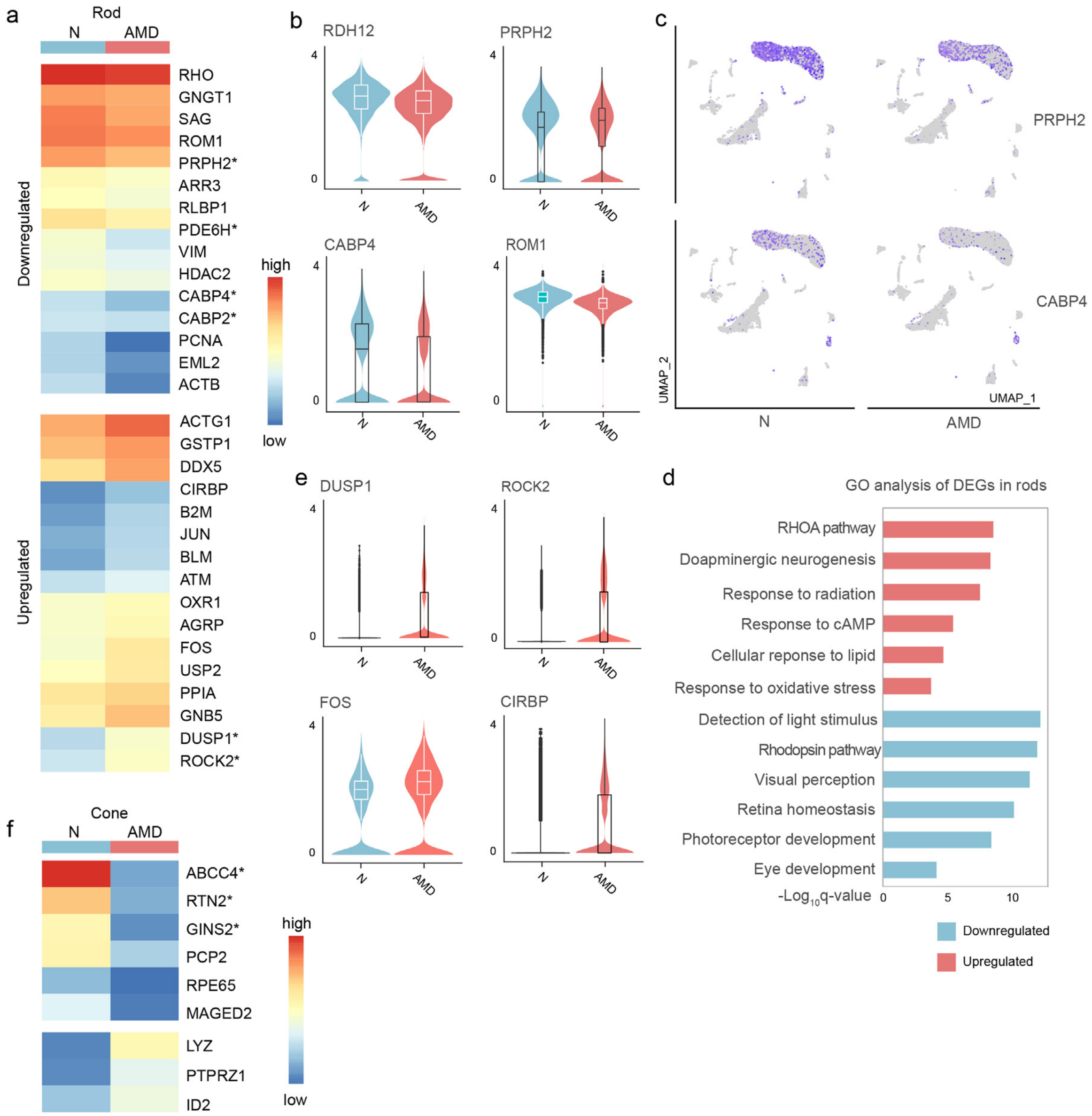

Fig. 4. Photoreceptors show distinct transcriptional alterations in spontaneous dry AMD. (a) Heatmap of the expression of DEGs in microglia from two groups of macaques. (b) Violin plots of *RDH12, PRPH2, CABP4* and *ROM1* in two groups of macaques. N, normal aged controls; AMD, spontaneous dry AMD macaques. (c) UMAP embeddings of *PRPH2* and *CABP4* in retinal cells in two groups of macaques. (d) GO analysis of upregulated/downregulated DEGs in rods. (e) Violin plots of *DUSP1, ROCK2, FOS, CIRBP* in rods in two groups of macaques. (f) Heatmap of expression of upregulated/downregulated DEGs in cones in two groups of macaques.

electroretinography (ERG) tests, which assess the electrical response of the retina to photic stimulation (Fig. 7d). The mice exhibited reduced a-wave and b-wave amplitudes compared to those in normal controls, and the ferroptosis inhibitor reversed these effects (Fig. 7d, e). These results support the potential of ferroptosis inhibitors as new treatment strategies for dry AMD.

4. Discussion

4.1. Global transcriptomic alterations of retinas with SD-AMD were exhibited

Several studies have implied prominent roles of chronic inflammation (especially the complement pathway), mitochondrial dysfunction, oxidative stress, lipid metabolism, and autophagy in dry AMD pathobiology, but they were mostly restricted to mouse models [2,30–32]. Genetic studies based on bulk RNA sequencing techniques have contributed to our understanding of the biological underpinnings of AMD, highlighting genes associated with the complement system and mitochondrial proteins [5,33]. With developments in scRNA-seq techniques, several studies have focused on transcriptomic analysis of the retina and choroidal vascular tissues of humans or NHPs, achieving with molecular resolution and providing insights into the retinal structures as well as laying a foundation from which researchers may begin to understand human retinal disease pathogenesis [6–8,34]. Voigt et al. revealed the cell type-specific regulation of cell cycle-associated genes in human RPE choroid tissues from donors with neovascular AMD [34]. Despite extensive investigations on retinal structures and neovascular AMD pathology, our understanding of the mechanisms underlying dry AMD at molecular resolution remains unclear, and no effective preventative or treatment measures are available for dry AMD.

In this study, we have reported a primate single-cell transcriptome landscape of retinas with SD-AMD, providing a detailed analysis of AMD based on diverse retinal cell types. Consistent with the current understanding and histological research, retinal tissues from the affected animals exhibited degenerative changes compared to those ob-

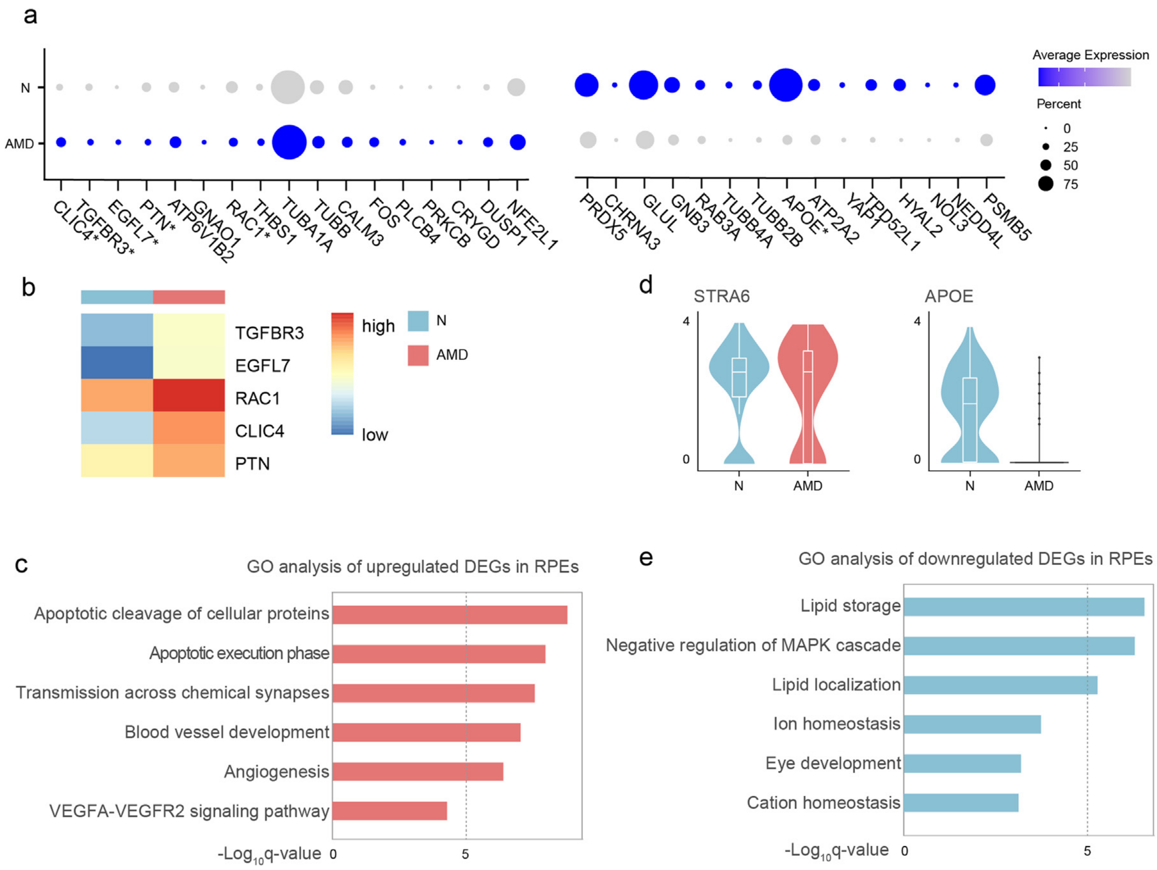

Fig. 5. Gene expression networks show cell-type-specific regulation among different cell types. (a) Dot plots of expression of upregulated/downregulated DEGs in RPE cells in two groups of macaques. (b) Heatmap of expression of upregulated neovascularization-associated genes in RPE cells in two groups of macaques. (c) GO analysis of upregulated DEGs in RPE cells. (d) Violin plots of expression of *STRA6* and *APOE* in RPE cells in two groups of macaques. (e) GO analysis of downregulated DEGs in RPE cells.

served in aged, unaffected animals [3]. Features of SD-AMD were associated with a generally more degenerative and dysfunctional transcriptomic landscape, suggesting increased oxidative stress, elevated response to stress, and global activation of the apoptotic signaling pathway [32].

4.2. Müller glia were shown to have an important role in AMD pathogenesis

Müller glial cells are important in maintaining the viability of photoreceptors and neurons, stabilizing retinal structure, and regulating retinal immune responses [12]. Researchers have discovered a remarkable retina regeneration capacity of Müller glial cells in zebrafish [35,36]. After retinal injury, the Müller glial cells of zebra fish were shown to go through a reprogramming event, exhibiting stem cell properties and generating progenitors for retinal repair. Unlike in rodents, mammalian Müller glial cells in vivo responded insufficiently to growth factor and cytokine stimulation, exhibiting weaker ability to produce new neurons, which is insufficient for retina regeneration [37]. Several studies attempting to enable mammalian Müller glia to acquire regenerative properties in vivo have found limited success [38,39]. Recently, Pollak et al. forced the expression of the proneural transcription factor Ascl1 in mouse Müller glial cells in vitro, enabling them to generate new neurons [40]. Primary human Müller glial cell cultures were reported to successfully produce photoreceptors and RGCs [41,42]. In this study, we identified a subtype of Müller glial cells expressing photoreceptor markers, including *RHO, GNAT1*, and *ROM1*, in the retinas of both normal aged and AMD macaques. Notably, SD-AMD macaques exhibited decreased cellular composition of the PR-like subtype as well as down-regulation of light sensitivity- and function-associated genes. Our results suggest the existence of a certain Müller glial cell subtype with the potential to transition into a distinct functional state; this could serve as the origin for photoreceptor progenitors after retinal damage. More importantly, this potential seemed to be weakened in SD-AMD macaques, suggesting that decreased proportions and weakened transition potential of this Müller glia subtype could contribute to AMD pathogenesis. Furthermore, we identified upregulated expression of KDR, which encodes VEGFR2 and is important in retina neovascularization in wet AMD pathogenesis. This result suggested that neovascularization-associated biological processes could take place even in early SD-AMD.

4.3. Expression of M2-type markers was upregulated in SD-AMD-associated microglia

Microglia may affect RPE and photoreceptor integrity, consequently promoting advanced neovascular AMD progression [43]. Prior research has highlighted the central roles of the innate immune system in AMD pathogenesis, such as the accumulation of complement components in drusen deposits, genetic inheritance of mutations associated with complement systems, and activities of cellular mediators of microglia [44]. In SD-AMD animals, microglia also appeared to participate in disease pathogenesis, exhibiting activated signatures. We identified upregulated expression of *C3AR1*, which was closely associated with complement system activation. Recent studies have found C3AR1 to be associated with neuroinflammation and to act as a regulator of microglia reactivity [45,46]. We hypothesized that microglia also participate in SD-AMD pathogenesis, with C3AR1 as the main mediator. More recently, researchers have observed microglia polarization to the pro-angiogenic M2 phenotype in neovascular AMD [14,44]. Here, we report the upreg-

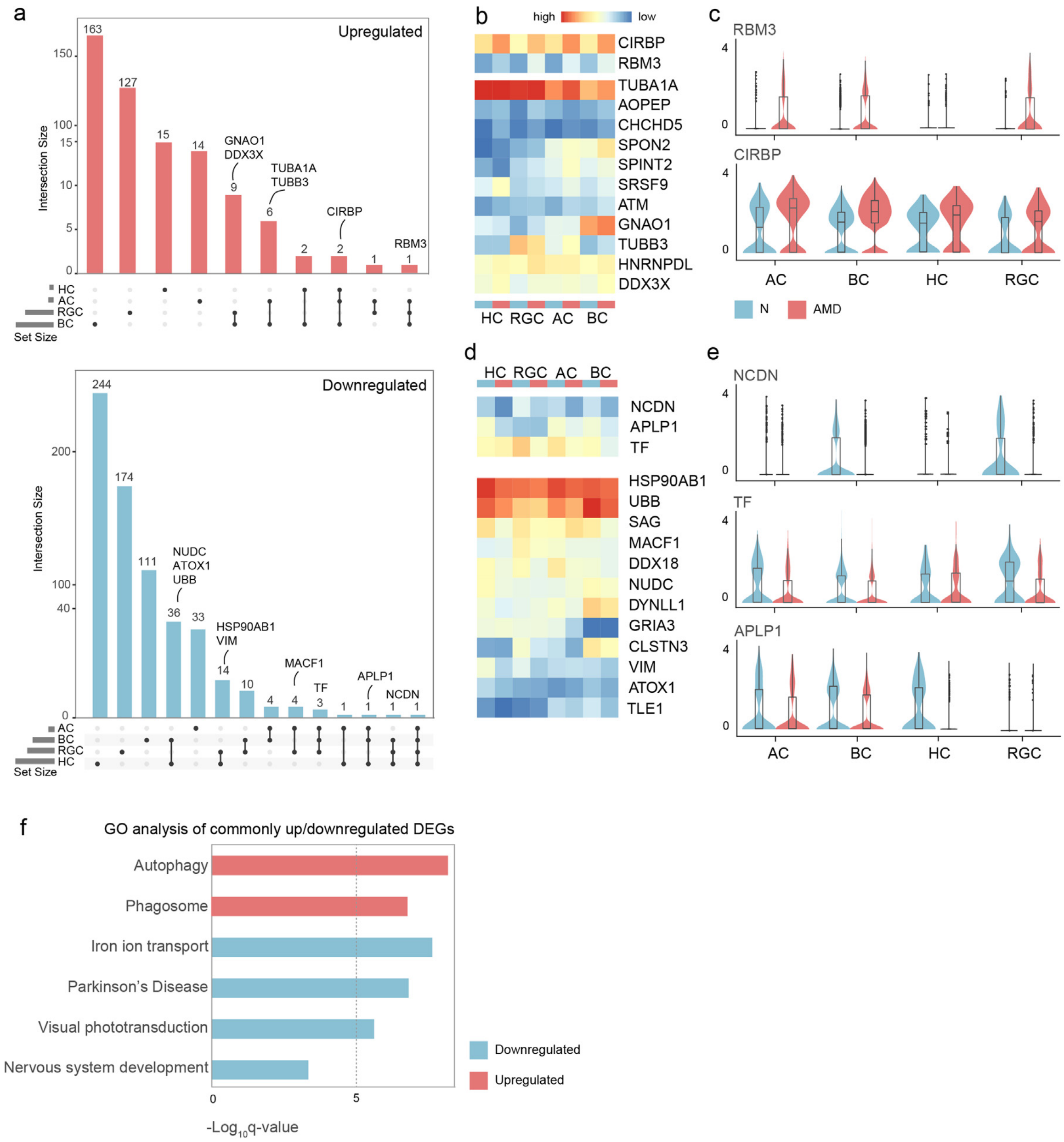

Fig. 6. Global alterations of dry AMD on neural cells. (a) Venn diagrams show the numbers of commonly upregulated/downregulated DEGs in four types of neural cells. Abbreviations for each type of cells are the same as that in Fig. 1. (b) Heatmap of expression of commonly upregulated ($n \geq 3$) genes in four types of neural cells. N, normal aged controls; AMD, spontaneous dry AMD macaques. (c) Violin plots of commonly upregulated genes (*RBM3, CIRBP*) in four types of neural cells in two groups of macaques. (d) Heatmap of expression of commonly downregulated ($n \geq 3$) genes in four types of neural cells. (e) Violin plots of commonly upregulated genes (*NCDN, TF, APLP1*) in four types of neural cells in two groups of macaques. (f) GO analysis of commonly upregulated/downregulated DEGs in four types of neural cells.

ulated expression of M2-type markers in SD-AMD-associated microglia, suggesting that M2 polarization continuously participates in both the early onset and the development of AMD.

4.4. RPE cells exhibited neovascularization-associated transcriptomic alterations

Clinically, dry AMD is defined by RPE and photoreceptor abnormalities [4]. Robust and consistent experimental evidence based on both human donors and animal models with AMD supports RPE defects as the main drivers of AMD pathology. Disrupted mitochondrial architecture in RPE cells, accompanying elevated oxidative stress and local RPE inflammation, has been highlighted based on experimental evidence and bulk RNA-sequencing [47]. As mentioned above, Voigt et al. performed scRNA-seq on human RPE-choroid tissues and highlighted the role of cell cycle-associated genes in neovascular AMD pathology [34]. We identified upregulated expression of neovascularization-associated genes, such as *RAC1, PTN*, and *EGFL7*, in SD-AMD RPE cells [25]. Upregulated *RAC1* expression has been suggested to activate HIF-1, upregulate VEGF production, and strengthen neovascularization in RPE cells, whereas *PTN* and *EGFL7* were found to encode proteins that reportedly participate in VEGF-induced angiogenesis [24,48,49]. Our results showed that neovascularization-associated transcriptomic alterations in RPE cells occurred even in early SD-AMD, indicating that both dry and wet AMD were associated with similar RPE dysfunction during early onset of the disease.

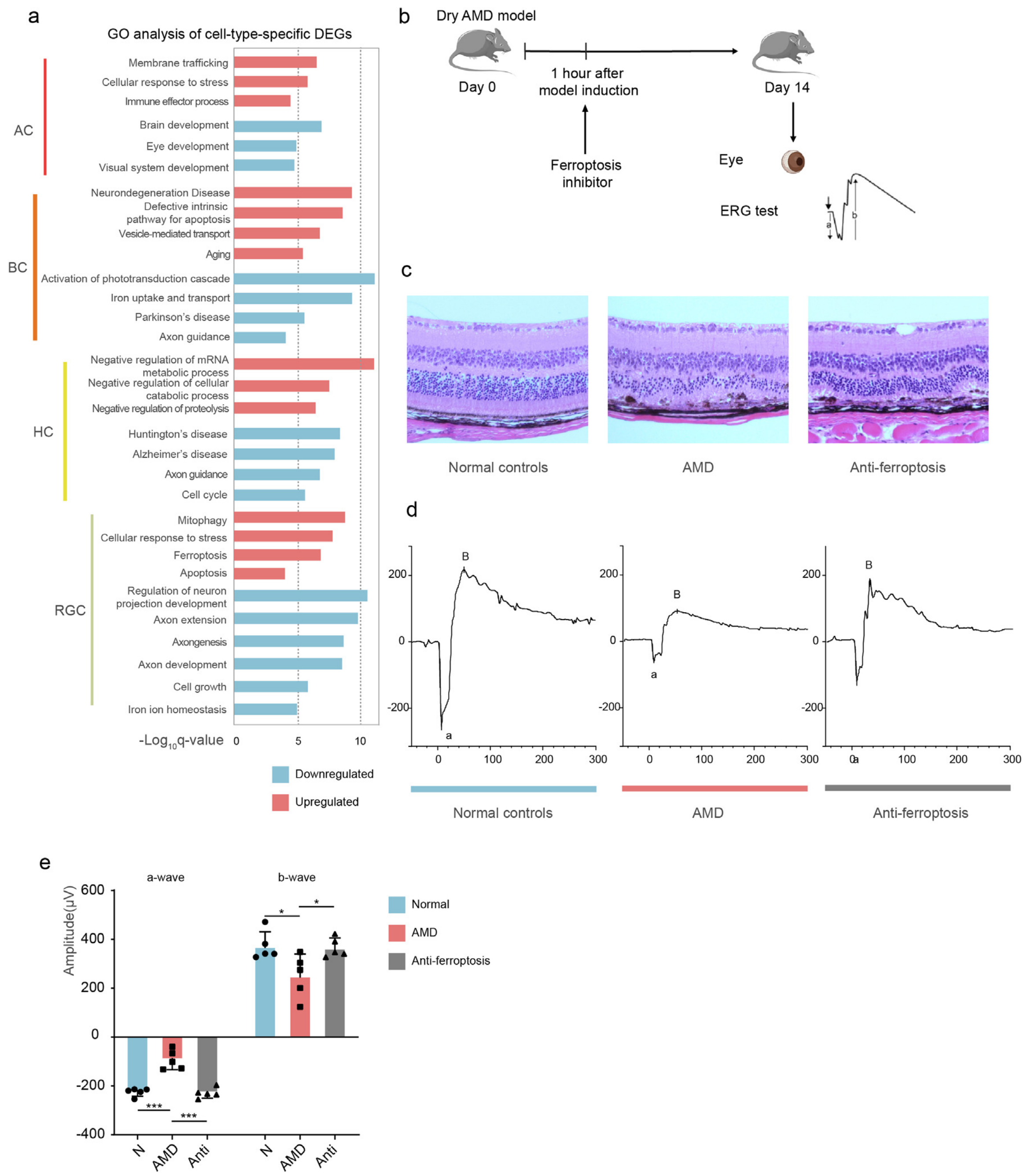

Fig. 7. Cell-type specific transcriptomic alterations in four types of neural cells suggested the roles of ferroptosis in dry AMD pathogenesis. (a) GO analysis of cell-type-specific upregulated/downregulated DEGs in four types of neural cells. (b) Research design. Dry AMD mouse models, ferroptosis inhibitor-treated dry AMD mouse models and normal controls were studied. (c) Images of the H&E staining of mouse retinal tissues in three groups of mice. AMD, dry AMD mouse models; anti-ferroptosis, ferroptosis inhibitor treated dry AMD mouse models. (d) Averaged scotopic electroretinograms of three groups of mice are shown. The averaged traces for three groups of mice are plotted. (e) Bar plots of a-wave (left) and b-wave (right) amplitudes in three groups of mice (five mice per group). Mean ± SD with individual values is presented. Data are sampled from three independent experiments (five mice per group). P-values are exact two-sided generated by students' t-test. N, not significant, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

4.5. Cellular response indicated that retinal neural cell lesions may occur in early AMD

The retina is a neuron-rich tissue containing over 60 distinct neuronal cell subtypes, each potentially displaying profound alterations in human neurodegenerative disease [50]. Although multiple clinical studies have identified the early onset of vision dysfunction in dry AMD patients, transcriptomic changes in neural cells associated with SD-AMD were previously unknown. Moreover, recent research based on large-scale multifocal electroretinography tests showed that the dysfunction of photoreceptors and bipolar cells occurred even in early SD-AMD, suggesting that neural cells were also affected [20,51,52]. In our study, we identified common changes across different types of neural cells, including an enhanced cellular response to oxidative stress, decreased cell survival, and neuron growth, based on transcriptomic features. These findings suggested that lesions of retinal neural layers occur in early AMD, resulting in vision dysfunction. We identified global upregulation of the expression of *APLP1*, which participates in AD pathogenesis, indicating potential common mechanisms underlying the two distinct degenerative diseases [28].

Prior studies have implicated iron in AMD pathogenesis, revealing increased iron levels in the retinas of patients with AMD [53,54]. Genetically modified mice that accumulate iron in the retina also exhibit retinal degeneration, further supporting the involvement of iron in AMD pathogenesis [55]. In this study, we similarly identified the widespread disruption of iron homeostasis in multiple types of neural cells; therefore, we conducted validation experiments by treating dry AMD mouse models with ferroptosis inhibitors.

Our study had some limitations. The sample size of the normal aged control group was smaller than that of the SD-AMD group. Furthermore, the overall numbers of samples were limited by the large-scale SD-AMD selection process. Finally, the current study did not include monkeys with late-stage SD-AMD owing to sample limitations, marking an area which warrants further research.

5. Conclusion

We established a primate-based single-cell transcription atlas of aged and SD-AMD retinal tissues. By characterizing cell type-specific transcriptomic signatures, we identified nine distinct cell types and annotated SD-AMD-associated DEGs. We further revealed common transcriptomic alterations associated with increased oxidative stress, activation of the apoptotic signaling pathway, and evident degenerative tendencies, as well as cell type-specific alterations in SD-AMD monkeys. The AMD-associated DEGs were also suggestive of cell type-specific alterations. Overall, our results provide retinal cell atlases for SD-AMD and normal aged crab-eating macaques, offering a valuable resource and revealing promising avenues for further research.

Declaration of competing interest

The authors declare that they have no conflicts of interest in this work.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (2017YFA0105804 and 2020YFA0112701).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.fmre.2023.02.028.

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